Three-output contraDC interferometer

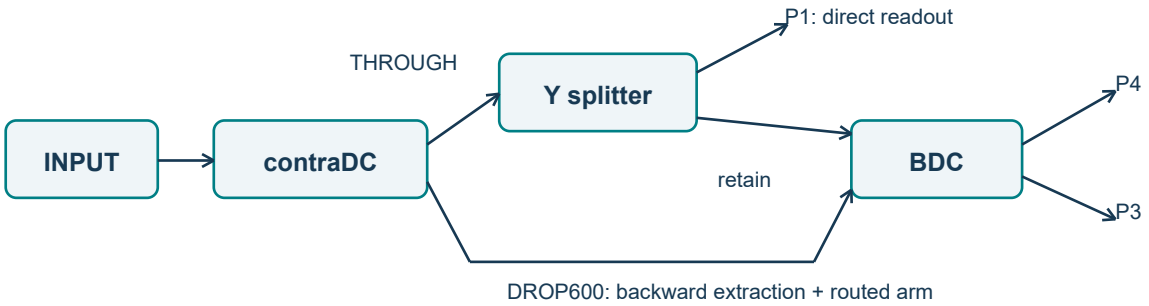
A passive silicon-photonic wavelength readout and advanced interferometric course project

Public edX username: weijiany | Platform: 220 nm silicon-on-insulator, quasi-TE | Status: pre-fabrication

Abstract

This design converts monochromatic wavelength into three output-power ratios while treating a common input-power scale as unknown. A contra-directional coupler (contraDC) creates THROUGH and backward DROP responses. A Y splitter provides one direct output and a retained arm; a broadband directional coupler (BDC) mixes that arm with DROP to produce the other two readouts. The main circuit uses one input and three output grating couplers (GCs). A separate two-GC path supports transmission-reference measurements.

Retained component finite-difference time-domain (FDTD) simulations support a sampled target DROP peak of 53.6237% at 1554.125 nm, near-balanced Y/BDC responses and conditional complex circuit calculations. These component values are not fiber-to-fiber efficiency. The final layout passes the executed complete basic design-rule check (DRC) and functional/design-for-test (DFT) check. No fabricated spectra, calibrated wavelength accuracy or yield data are available.



Functional diagram only; actual routing and the independent two-GC reference are shown on page 2.

Course objective and scope
The course requests waveguide group-index extraction[1]. Appendix A1-A3 tests the current-device calibrated inverse in 24 cases, with 73 fit and 77 held-out wavelengths. Known module phase and relative gain are required; unknown device delay remains degenerate with ng. The55-68 nm local FSR proxy does not establish the approximately 50 nm measurement-window requirement. The straight-MZI control is not in the mask.

The present report replaces the previous reference/MZI/ring design description. It documents the current advanced circuit and its limitations rather than implying that the old peer review evaluated this changed design.

1. Architecture and final mask

One connected four-GC main circuit plus a separate two-GC transmission reference

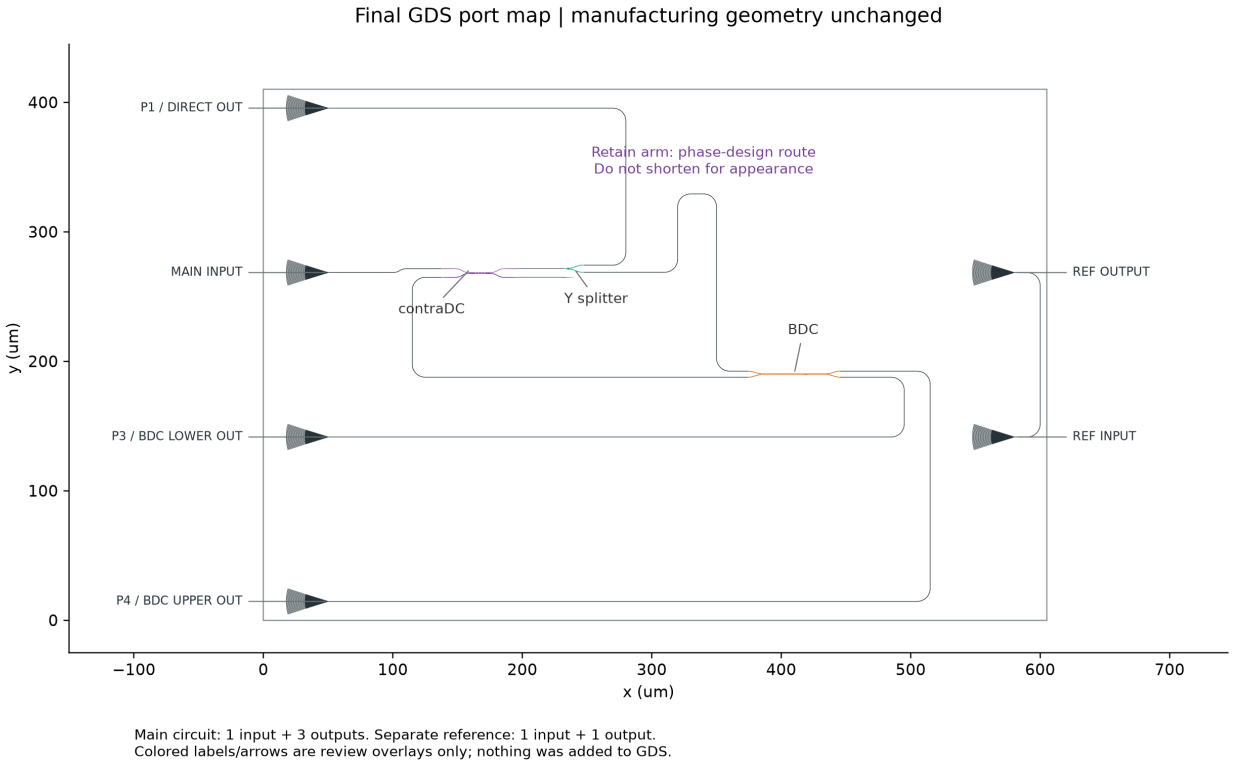


Figure 1. Actual current-GDS port map. Colored labels and arrows are explanatory overlays, not added fabricated geometry.

Port / channel	GC center (um)	Role
GC2 / IN	(50,268.5)	Only main laser input
GC1 / P1	(50,395.5)	Y direct/upper branch
GC3 / P3	(50,141.5)	BDC lower opt4 output
GC4 / P4	(50,14.5)	BDC upper opt3 output
REF_IN / REF_OUT	(580,141.5)/(580,268.5)	Independent path, excluded from main q sum

The floorplan is 605 um by 410 um. Main-array pitch is 127 um. The course testing page describes three detector fibers around one input per connected circuit[1]; it does not impose a four-GC total for the whole file. All GCs face right. Existing opt_in labels, library cell origins and contents, PinRec and DevRec records are preserved.

The latest text-only layout polish leaves all physical geometry unchanged. Current GDS size is 217248 bytes, top cell EBeam_weijiany and database precision1 nm. Engineering annotation layers must not be exported as extra silicon.

2. Waveguide model and simulated mode

Mode profile and finite-model dispersion are different retained numerical references

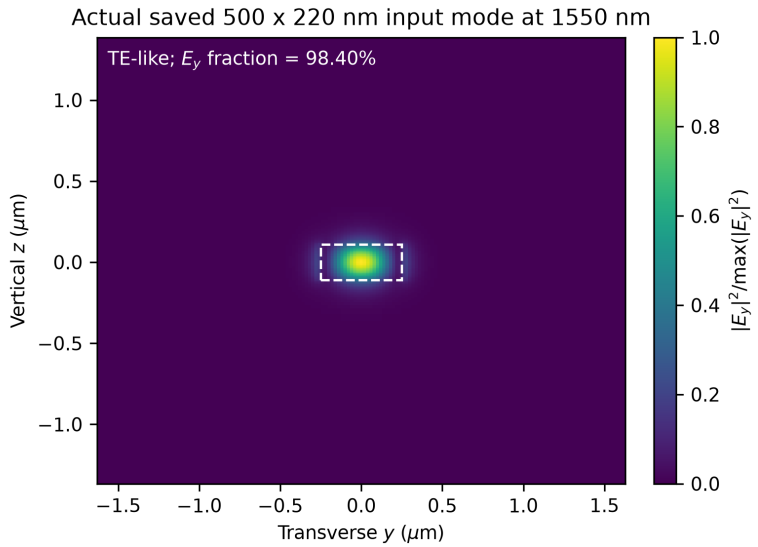


Figure 2. Actual Y-input 500 nm mode at 1550 nm, one retained single-frequency native field slice. Intensity is normalized for display only. This is not the two-line L3 reference used for the dispersion polynomial.

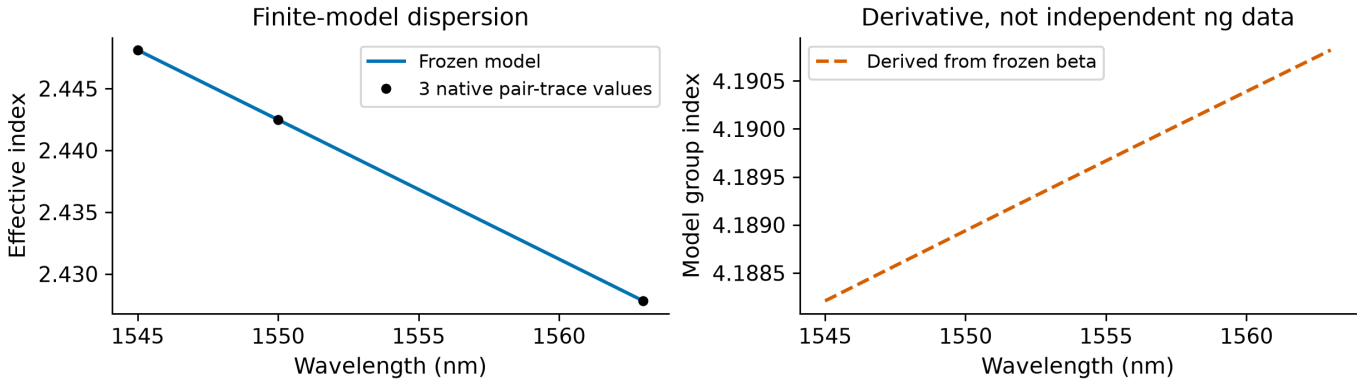


Figure 3. Retained three native L3 pair-trace effective-index samples with a frozen quadratic beta model; ng is its derivative, not an independently solved or measured spectrum. Interpolated lines stay within 1545-1563 nm.

x=(f -193.414489032258 THz)/(1 THz)

beta(f)=[9.900982796830407 +0.08779368785155338x -0.0000122558817002169 6x^2] rad/um

$n_{eff}(f)$=c beta/(2 pi f); $n_g(f)$=[c/(2 pi)] d beta/df = n_{eff} -lambda d n_{eff}/d lambda

Equivalently, for u=$lambda_{um}$-1.55, the derived second-order compact form is n_{eff} approximately 2.442475048 -1.126751502u -0.047077241u^2. This is a Taylor representation of the same frozen beta model, not another fit.

Use c=299792458000000 um/s and frequency in Hz in the derivative. The L3 finite-PEC reference contains a near-degenerate two-TE subspace, not a uniquely certified isolated-guide mode. Three fit nodes and one held-out beta check do not establish derivative order, mesh convergence or process-lot accuracy. Center model ng is 4.188940; the Y single-mode profile must not be used to erase this model distinction.

3. Group index from fringe spacing

A reproducible numerical control tests the inverse; it does not add a new fabricated MZI

For two otherwise identical straight arms with known unequal length DeltaL and frequency-independent component phase, the differential phase is DeltaPhi=beta(omega)DeltaL+phi0. The normalized ideal-MZI output is I=(1+cos DeltaPhi)/2. Successive equal-phase maxima differ by 2pi, giving:

Delta_tau=d DeltaPhi/d omega=n_gDeltaL/c;
FSR_f approximately 1/|Delta_tau|;
n_g approximately c/(|DeltaL| FSR_f) approximately $lambda^2$/(|DeltaL| FSR_{lambda}).

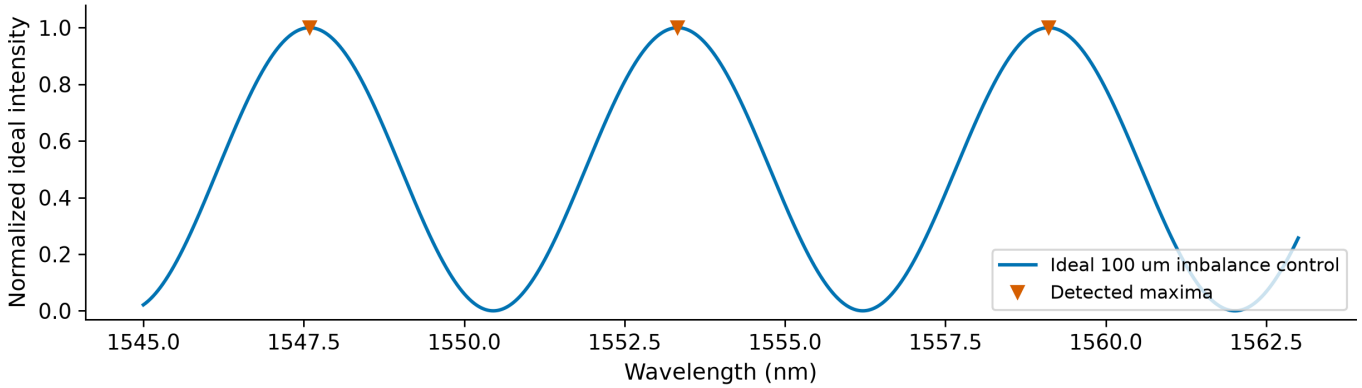


Figure 4. Ideal straight-MZI control built from the frozen beta polynomial. The control DeltaL=100 um and phi0=0 are numerical assumptions, not the current-mask arm lengths. Peak extraction uses fixed0.001 nm sampling within 1545-1563 nm; no dispersion extrapolation or new EM solve.

Peak interval (nm)	Frequency FSR (THz)	Recovered model ng
1547.589278 to 1553.327903	0.715664903	4.189006007
1553.327903 to 1559.108087	0.715521950	4.189842923

The two peak intervals give model ng=4.189006007 and 4.189842923 at their interval reference frequencies. Agreement with the derivative of the same quadratic model is better than1.1e-10 in absolute ng. This tests formula, peak sampling and bookkeeping only; it is not independent electromagnetic validation or physical precision.

Experimental use of this equation
Measure a wavelength-calibrated spectrum, identify consecutive fringes under a slow envelope, determine DeltaL from the same physical reference planes, and propagate calibration/length/model uncertainty. This direct inverse is conditional on negligible differential component dispersion. A 1 pm laser sampling step is not a 1 pm accuracy certificate. The present advanced circuit needs the composite model on page 5.

4. Composite transfer and phase ledger

The current circuit contains dispersive modules, unequal bend counts and explicit reference planes

Let t(lambda) and d(lambda) be the core/interface target complex responses after the declared common-basis/reference-plane transformations. Let y_u,y_l be the Y transfers; C(lambda) is the retained simulated target 2-by-2 forward BDC response. The factors g include only the corresponding physical propagation, losses and explicitly counted bend-coupon phases. In the declared forward model:

$a_{direct}=g_1\, y_u\, g_{TY}\, t$;
$a_R=g_R\, y_l\, g_{TY}\, t$; $a_D=g_D\, d$;
$[b_{P4}, b_{P3}]^T=\mathrm{diag}(g_4,g_3)\, C\, [a_R,a_D]^T$.

For output j, the interference term is $2\mathrm{Re}(C_{jR}C_{jD}^{*}\, a_R a_D^{*})$. Its phase derivative contains both arm and BDC dispersion. Power is evaluated with the retained tangential E/H flux metric; a scalar coefficient modulus is not silently substituted for every raw-profile normalization.

Length / convention	Actual retained master value
THROUGH-to-Y	36.114346000 um
Retained arm from Y	307.363949757 um
DROP arm	350.879949743 um
RETAIN minus DROP label	-43.515999986 um; only this pair of paths
Full external geometric difference	-7.401653986 um; includes THROUGH-to-Y
R10 coupon arclength reference	20.708173212 um per coupon
Coupon-consistent effective difference	-48.818000409 um; phase bookkeeping, not a physical straight-MZI DeltaL

The retain cap was shortened by 63.516 um relative to the previous mask while preserving endpoints, bend count and component geometries. Phasors use exp(-i omega t), with explicitly oriented E polar vectors and H axial vectors. Mode gauge, virtual leads and reference-plane propagation are tracked before connection; neither complex conjugation nor an adjustable constant phase repairs a missing bridge.

What the mask can identify
A composite differential delay can be estimated under a stated calibrated device-phase model. FSR alone does not separate waveguide ng from contraDC/Y/BDC/bend dispersion. The independent two-GC reference is one path, so it supplies no interferometric FSR by itself.

5. Actual component simulation evidence

Target-channel powers use actual main incident-mode normalization; component evidence is reused

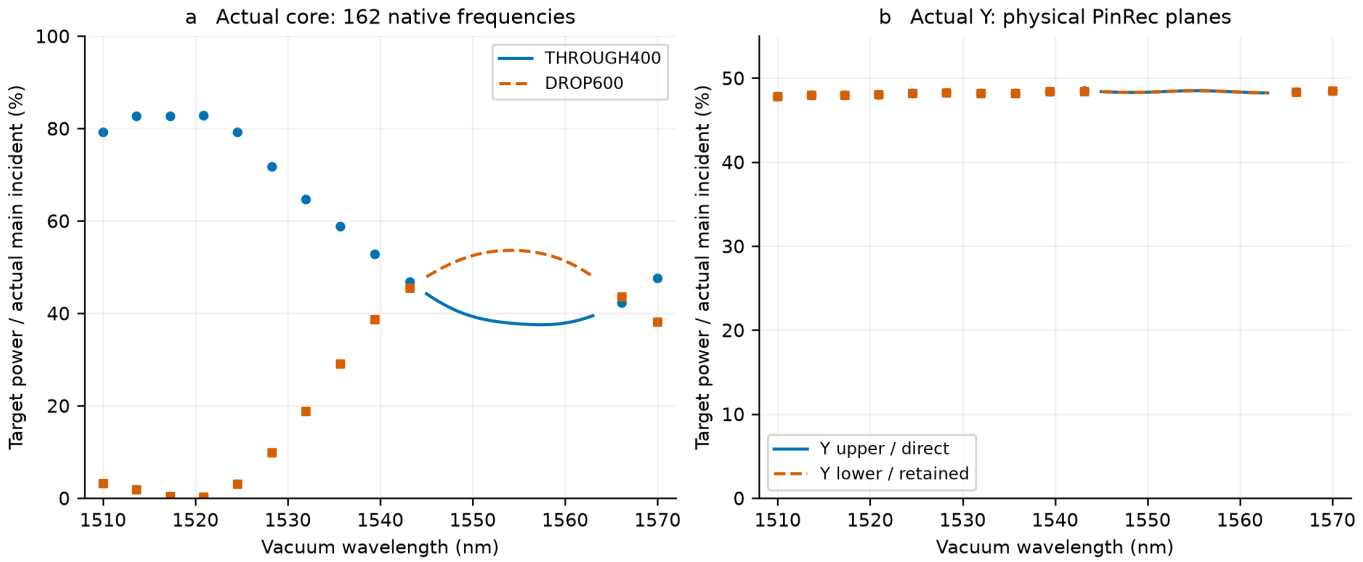


Figure 5. Retained core and Y target responses at native frequencies. Broad-band markers and the dense central1545-1563 nm segment have different sampling densities. Curves are not independent wavelength interpolation validation.

Component / condition	Retained observation	Interpretation
contraDC400 input	DROP sample peak 53.6237% at 1554.125 nm;52.8714% at 1550.744 nm	Core target extraction; not whole-chip efficiency
Matched 400/600-to 500 interface	Near 1550.744 nm net spatial T99.5026% /99.5849%	Two actual source-forward responses, not full S
Y, near 1550.744 nm	Each branch approximately 48.35%	Actual near-balanced target split, not ideal lossless
BDC upper input	Target outputs 50.008% /49.641%	Keep complex responses and output order
BDC lower input	Target outputs 49.116% /50.093%	Same stored target basis; residuals remain
R10 bend coupon	Target forward 99.846% near 1550.744 nm	Its phase and reverse-transfer assumptions matter

The nominal core/Y/BDC/R10 cases have 162 native frequencies, including 145 points spaced 0.125 nm within 1545-1563 nm. The matched interface retains 17 native frequencies and is explicitly interpolated in dense circuit calculations. A dense circuit curve is therefore not a whole-chain 162-frequency native EM solve.

Small unintended source-mode components, finite modal apertures, residual modes/radiation and normalization conventions are retained in the analysis. A target-channel coefficient is not a mathematical pure-unit-input complete scattering matrix. Native low late-time energy and a near-unity throughput are not independent mesh, time and boundary convergence tests.

6. Conditional system spectrum and delay

Fixed final route; actual component caches; no fitted phase to force a desired result

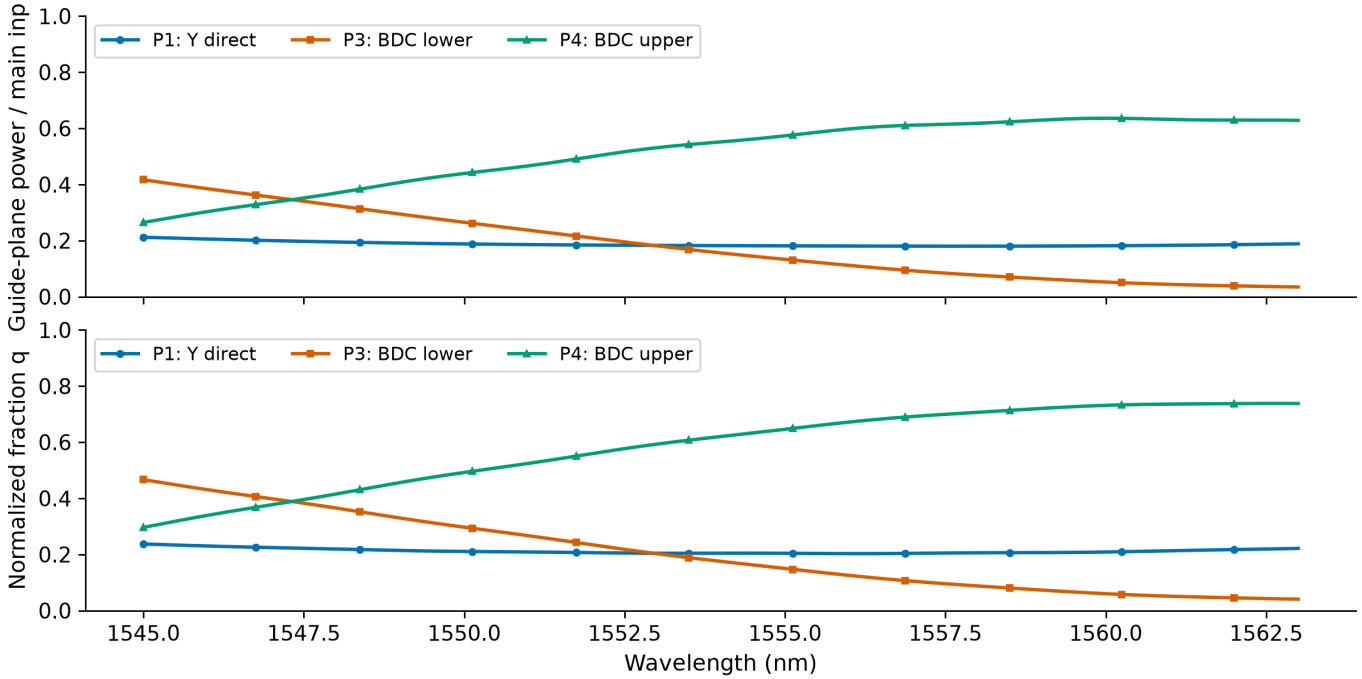


Figure 6. Current fixed-GDS conditional forward response. Main-output q is defined at the stated guided reference planes before GC coupling and detector gain. Lines contain declared interface interpolation and the finite beta model; they are not measured spectra.

$q_j(\lambda)=P_j/(P_1+P_3+P_4)$, $j=1,3,4$;
$\Theta_j=\arg(a_D/a_R)+\arg(C_{jD}/C_{jR})$;
$\Delta\tau_j=d\Theta_j/d\omega$; $FSR_{f,j}$ approximately $1/|\Delta\tau_j|$ only for slow envelopes.

In this one retained flux/L3 scenario, input differential delay spans -139.06 to -127.86 fs. Including the BDC interference phase gives local phase-slope FSR proxies of about 55-68 nm. The plotted18 nm window does not directly observe those complete phase cycles, and amplitude-envelope extrema are not automatically phase fringes. The reassembled powers/q match the retained baseline to 1.29e-13; that is a computation-consistency check, not independent physical validation.

Wavelength-readout limitation
Common input scaling cancels in q; independent GC/detector gains, background, temperature and mode changes do not. Different admitted propagation-phase models previously changed an individual q by up to 22.096 percentage points. That is model sensitivity, not a measured fabrication distribution or failure probability. An old conditional 6.7pm decoder result must not be carried over as final precision.

Appendix A1-A3 supplies a calibrated current-device ng fitting method and 24 same-model recovery tests using73 fit and 77 held-out nodes. This extends the straight-MZI control to the actual composite topology. Unknown device delay, relative spectral gains and alternative phase branches still limit physical ng identification; no uncalibrated or globally unique measurement is claimed.

7. Return risk and validation boundaries

Actual reverse-source cases constrain important errors without constituting complete all-port validation

Retained reverse-source case	Target TE return in 1545-1563 nm	Scope
Y retained branch	0.1076%-0.80385% on 150 nodes	Local reverse illumination; finite actual source mixture
Central THROUGH readout side	0.00683%-1.92839% on 5 anchors	Actual 4 ps case; matched central stored basis
Central DROP readout side	0.51892%-2.07598% on 5 anchors	Actual 4 ps case; unknown central blocks still matter

These are target responses under reverse illumination, not main-input insertion loss or a manufacturing-yield estimate. Actual forward/reverse target fields, power direction and reference planes were checked before a bounded feedback calculation. The unknown central return block was varied within declared passive reciprocal completions; external GC/source loads were assumed matched for those scenarios.

Existing bounded test	Observed result	What it does not prove
Two Y/central conditional feedback families	Maximum q change 0.32146 percentage points in the selected baseline	No universal feedback bound
1728 finite route-phase stress scenarios	Maximum q change 0.83352 percentage points	Not all unknown blocks times all phases; not fabrication statistics
Y conditional-model comparison	Maximum q difference 0.21377 percentage points	Not complete module uncertainty

The latest feedback evidence did not justify another physical layout change in this engineering review. It also does not erase the earlier propagation-phase sensitivity. These are separate conditional comparisons, not competing estimates of one certified error bar.

Convergence and completeness
There is no whole-chain 3D-FDTD result, complete guided/radiative full S, or final mesh/time/domain convergence certificate. Windows coarse1-to 2 ps output changes were small, while Linux output fields remained PML-sensitive. These solver/model results are kept separate from the cloud component evidence; no unrelated runs are relabelled as a single convergence study.

8. Fabrication checks and measurement plan

Local geometric correctness is established; process acceptance and experimental calibration remain distinct

Final-GDS check	Actual result / retained risk
Executed complete basic DRC	0 items; original 70 nm minus 1 nm tolerance rule body, IO signature adapted only
Official functional/DFT check	0 items;19 components and 18 nets
Mask and hierarchy	605x410 um;1 nm database; original library cells/opt_in preserved
Core	400/600 nm strips;336 nm pitch,25 periods;104 actualSi rectangles;80 nm minimum gap
Unused-core termination	10 um taper ending at 70 nm drawn width; nominal minimum, little CD margin
Stack assumption	220 nmSi,2 umBOX,oxideclad ideal model; actual lot sidewall/CD/etch acceptance not established by DRC

Planned experiment

Launch quasi-TE light into GC2 and record P1/P3/P4 simultaneously in the channel order on page 2. Start in 1545-1563 nm, within the course typical GC bandwidth. Measure dark/background and per-channel gain, verify detector linearity and avoid saturation. Convert dBm to linear power before forming q. Scan the separate REF path to assess path/coupling spectra; do not assume it cancels unequal losses of all four main GCs.

Record chip ID, temperature, polarization setting, input power, wavelength reference and repeated scans. Repeat at changed input power to test the common-scale assumption. Use a training wavelength scan and separate validation scans for a per-chip decoder; report distant aliases as failures rather than quoting only a favorable local noise result.

For ng extraction, calibrate relative spectral gains and module phases, then fit the composite response using the method in Appendix A1-A3. Its24 tests use73 fit and 77 held-out nodes. Unknown differential device delay cannot be separated from ng without calibration. No experimental result is included because the device has not been fabricated.

Fabrication review requested
Please review acceptance of the custom 80 nm coupled-core gap and 70 nm taper endpoint under the selected fabrication process. The drawing passes the executed rule deck, but ideal vertical-sidewall simulation and a1 nm database are not a guarantee of printability, optical yield or1 nm process accuracy.

9. Conclusions, course alignment and sources

Current-design document; explicit boundaries for later fabrication and data analysis

The submitted mask implements a physically connected one-input/three-output advanced interferometer and independent reference. Retained core, interface, Y, BDC, bend and reverse-source simulations support its intended signal path and bounded conditional modelling. The final text-only polish preserves the optical mask. The next evidence comes from fabrication and calibrated spectra, not from treating a nominal component throughput as a complete-system certificate.

Course report item[1]	Location / status
Geometry, polarization and mode	Pages 2-3 and 9; actual Y500 field plus finite-model dispersion distinguished
neff/ng and compact waveguide model	Page 3; ng is a finite-model derivative
Transfer function, parameters and spectra	Pages 5-7; actual mask lengths and conditional phase bookkeeping
ng-from-FSR derivation/test	Page 4 control; Appendix A1-A3: calibrated current-device method,24 tests; physical ng calibration pending
Experiment and analysis	Page 9 is a plan; no fabricated result or precision claim
Peer-review/design revision	Old submission was reference+MZI/ring; current advanced design is a substantial revision, not re-reviewed optical approval

[1] UBCx Phot1x,3T2026, Design Proposal - Report; Layout requirements for automated testing; Final Layout Submission. Authenticated course text read 17Oct2026. edX course.

[2] SiEPIC, openEBL-2026-10, submission/process instructions and checking source, inspected 17Oct2026. Official run repository.

[3] SiEPIC EBeam PDK and Library; retained installed rule deck and component definitions. Official PDK repository.

[4] Flexcompute, Tidy3D documentation; retained solver datasets report version 2.12.0. Official documentation.

[5] Project simulation records and analysis,15-17Oct2026: matched interfaces, core 162/Y/BDC/R10 nominal runs, phase-reference controls and three return-risk cases. Exact source paths, hashes, CSVs and scripts are indexed in the accompanying local source manifest.

[6] T.Kassis, V.Agarwal, Y.He, D.Patel and A.M.Brueckner, Scientific Agent Skills: A Library of Procedural Knowledge for Research Agents,2026, arXiv:2609.00065 (metadata checked17Oct2026). DOI. Used as analysis/figure workflow guidance, not optical evidence.

Preparation disclosure: AI-assisted coding, organization, drafting and internal review were used. Scientific figures derive from retained solver data or explicitly labelled analytic controls. This is a design report, not a claim of experimental measurement, complete phase closure, full S or manufacturing certification.

1. A concrete extraction method

Use the existing interferometer and three power spectra; do not substitute an ideal MZI that is absent from the mask

Result
The current topology supports a model-calibrated group-index inverse. Its finite numerical recovery test passes. Without a calibrated differential device-phase slope, intrinsic waveguide group index is not uniquely identifiable.

The THROUGH signal passes through the Y splitter into retained arm R. The backward DROP signal follows arm D. Their coherent fields mix in the broadband directional coupler (BDC), producing P3 and P4. The direct Y output P1 supplies an intensity reference. All sources below are retained native-component results and their existing phase/flux connections; no new electromagnetic (EM) solve is involved.

$P_j(w) = A(w)\, G_j(w)\, |C_{jR}(w)\, a_R(w) + C_{jD}(w)\, a_D(w)|^2$; j = 3,4.
$q_j = P_j / (P1 + P3 + P4)$.

Correct dark/background and relative channel response before forming q. The common laser-power scale A cancels; wavelength-dependent relative GC/detector gains do not. The independent two-GC reference alone does not determine every main-channel gain.

Separate propagation and calibrated device phase

$\mathrm{Theta}(w) = \mathrm{Psi}_{device}(w) + D\ \mathrm{beta}(w) + \mathrm{phi0}$;
$n_g(w) = (c / D)\, [d\mathrm{Theta}/dw - d\mathrm{Psi}_{device}/dw]$.

Here w denotes angular frequency. The current coupon-consistent model gives D = 48.818000409 um: retain uses four calibrated bend coupons and DROP uses two. This is the beta-sensitivity length after that declared accounting, not the full geometric arm difference (-7.401653986 um). The raw coupon transfer and all gauges/reference planes must remain fixed consistently.

Fit the measured observables, rather than first taking atan2

Use $\beta(f)=b0+b1\,x+b2\,x^2$, $x=(f-f0)/10^{12}$, f0=c/(1550 nm). Fit b1 and a constant phi0 to calibrated q3/q4 over wavelength; b0 is absorbed into phi0. Then $n_g(1550\ \mathrm{nm})=c\ b1/(2\pi\ 10^{12})$ with c in um/s. A second test also fits b2. Freeze the component amplitude and device-phase spectra independently of these propagation parameters.

A single-wavelength phase inversion is poorly conditioned: the two BDC outputs are nearly complementary. Across 150 wavelengths, the independently checked local Jacobian is full rank when only ng, constant phase and two constant relative gains vary. This supports the calibrated fit; it is not a proof of global uniqueness or real-chip calibration.

2. Verification using retained responses

Synthetic known-parameter recovery, training/holdout separation and a negative control

Test	Actual offline result
Nominal model	n_g(1550 nm) = 4.188939875; finite L3 two-guide TE-pair trace proxy
Known curvature	15 injected (ng, phi0) cases; ng = 3.8, 4.0, 4.18894, 4.4, 4.6; three phases
Curvature also free	9 injected cases; all three parameters recovered
Parameter fitting	73 even uniform nodes; 72 odd nodes + 5 native anchors held out
Checks	Baseline q recomposition <= 2.78e-16; compensated gauge change <= 3.34e-16

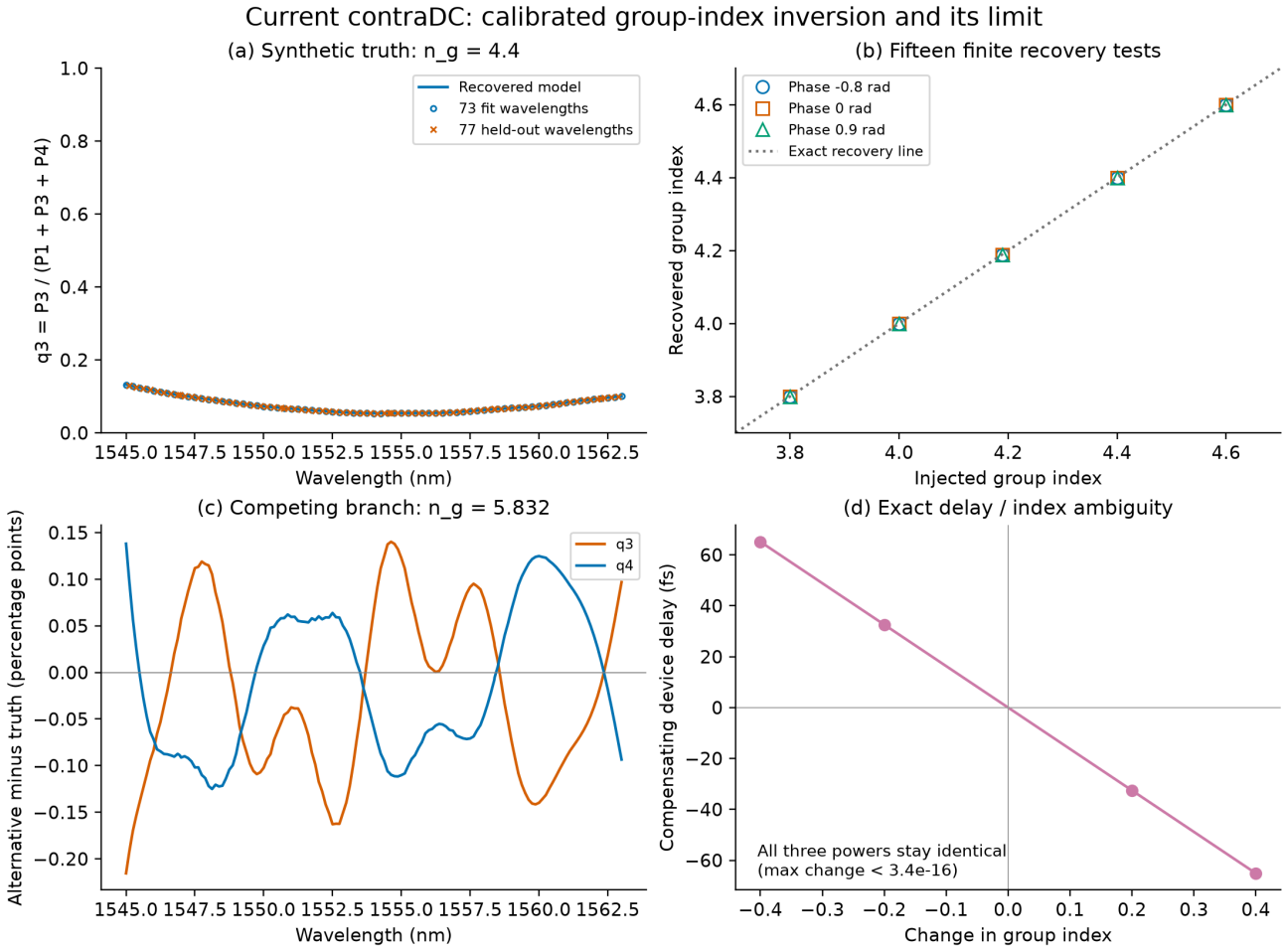


Figure 1. (a,b) Known-parameter recovery in the fixed calibrated model. Recovered/target markers overlap. (c) A competing local branch remains at ng about 5.832. (d) Allowing a compensating unknown device delay gives exactly identical powers. These are model tests, not new EM or chip measurements.

All 15 two-parameter and 9 three-parameter tests recover the injected targets to floating-point precision, with held-out normalized-power residuals below 1e-12. This is an algorithm/identifiability consistency result. Calibration at the held-out wavelengths still comes from the same supplied component model, so the holdout is not an independent physical validation.

3. What is proved, and what remains

The method is conditional; the exact missing calibration can be named

An exact counterexample to uncalibrated extraction

beta(w) -> beta(w) + delta_n_g(w-w0)/c;
Psi_{device}(w) -> Psi_{device}(w) - D delta_n_g(w-w0)/c.

This transformation leaves Theta and every output power unchanged. In the actual cached model, increasing ng by 0.2 and adding -32.567864 fs of differential device delay changes q by no more than 2.3e-16. The augmented Jacobian loses rank. More wavelength samples cannot break this exact ambiguity without an independent constraint.

Uncalibrated device delay	Equivalent group-index bias
1 fs	0.0061410
5 fs	0.0307051
10 fs	0.0614102

The actual retained alternative projection/interpolation models give fitted ng values from about 4.18605 to 4.19534 when interpreted with the fixed primary calibration. This finite model comparison is not a confidence interval or a complete error bound. A nominal competing phase branch near ng=5.832 differs by only about 0.22 percentage points in q across the window, so realistic calibration/noise must be considered before claiming global branch selection.

An implementable experimental procedure

1. Scan P1/P3/P4 at known wavelengths, remove background and calibrate relative channel spectra.
2. Supply independently justified component phase versus wavelength, with the same mode gauges, material model and physical reference planes.
3. Fit ng, constant phase and any explicitly allowed gain/dispersion nuisance parameters; inspect competing branches.
4. Check held-out wavelengths and compare against the waveguide simulation. Report calibration sensitivity and device-phase uncertainty. Do not label the fitted result a direct uncalibrated measurement.

Course boundary
A partial-fringe calibrated spectral fit can estimate ng without observing a full free spectral range (FSR). That mathematical fact does not by itself satisfy the course design instruction that FSR should fit inside the approximately 50 nm test bandwidth. Current retained local phase-slope proxies are about 55-68 nm over an 18 nm supported window; no complete fringe-spacing measurement or teacher waiver is established here.

Sources: frozen-test-protocol.json; actual-calibrated-ng-inverse-test-results.json (results-v2); actual-power-observability-challenge-v1.json; conditional-ng-observability-proof-v1.txt. All source paths and SHA256 values are included in the accompanying reproducibility bundle. No fabricated chip spectra, physical precision, full scattering matrix or manufacturing yield is claimed.

Workflow reference: Kassis T, Agarwal V, He Y, Patel D, Brueckner AM (2026). Scientific Agent Skills: A Library of Procedural Knowledge for Research Agents. doi:10.48550/arXiv.2609.00065. Used for evidence/claim review, not as an optical source.